

BENJAMIN RAINS

DATA ENGINEERING AND ANALYTICS ENGINEERING

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PROFESSIONAL SUMMARY

Data engineer and analytics engineer with 2+ years building and operating healthcare data pipelines, warehouses, and analytics systems. Combines SQL, Python, dbt, Airflow, and AWS with dental business experience to turn complex clinical and financial data into tested models and useful reporting. Background in regulated scientific analysis brings a disciplined approach to data quality, validation, and root-cause investigation.

TECHNICAL SKILLS

Languages and analysis: SQL, Python, Pandas, Jupyter | Warehousing and modeling: dbt Core, PostgreSQL, MySQL/MariaDB, Snowflake, dimensional modeling, incremental ELT, data reconciliation | Cloud and pipelines: AWS S3, EC2, RDS, Lambda, EventBridge, Apache Airflow, Airbyte, Docker | Development: Git/GitHub, FastAPI, REST APIs, automated testing, documentation, monitoring

PROFESSIONAL EXPERIENCE

Analytics Engineer / Data Engineer | Merrillville Dental Center | Remote | 2024-Present

- Build and operate a healthcare data platform that replicates 432+ OpenDental tables into PostgreSQL staging, intermediate models, and dimensional marts for patient, treatment, scheduling, insurance, and payment analytics.
- Developed a metadata-driven Python/SQL ELT framework with schema discovery, incremental loading, deduplication, type enforcement, and restart-safe execution; improved core nightly replication runtime 5-10x.
- Develop and maintain 150+ dbt models and 3,700+ automated tests and controls covering keys, relationships, duplicates, financial reconciliation, and business logic.
- Built payment and insurance reconciliation that identified and resolved 62,000+ previously unmapped records, strengthening financial and claims reporting.
- Orchestrate nightly Airflow workflows for source refresh, ELT, validation, dbt builds, publishing, and notifications; use failure controls to prevent incomplete loads from propagating downstream.
- Translate stakeholder needs into metric definitions and reusable SQL/Python logic for treatment journeys, patient behavior, collections, and clinical operations.
- Deploy AWS-hosted workloads and build FastAPI data services; modernize ingestion and warehouse workflows with Airbyte and Snowflake.

Owner / Partner | TB Dental Holdings | 2017-2023

- Partnered with clinical and business stakeholders on dental-practice operations; translated scheduling, patient flow, provider productivity, insurance, and revenue workflows into KPIs and data requirements.

Biochemist Analyst II-III | Eurofins PSS at Eli Lilly | 2013-2021

- Investigated GMP/FDA-regulated production data and deviations using data-integrity controls, technical documentation, root-cause analysis, and disciplined change control.

FEATURED PORTFOLIO PROJECT

Healthcare Analytics Platform | dbtdentalclinic.com

- End-to-end ingestion, transformation, dimensional modeling, and analytics across 1M+ procedures and 1.3M+ payment allocations, with documented lineage, validation, and operational recovery workflows.

EDUCATION

MS, Data Analytics | Purdue University Global | 2025

BS, Biochemistry | Purdue University | 2010